EK-SMS Complete System Plan

Document Purpose

This document provides a comprehensive breakdown of the EL-KENDEH Smart School Management System, mapping every feature to implementation phases, and extracting a focused MVP.

PART 1: COMPLETE SYSTEM ARCHITECTURE

System Modules Overview

The complete EK-SMS consists of **7 core modules** containing **47 distinct features**.

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│                EK-SMS COMPLETE SYSTEM                │
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│                              │
│  ┌──────────────────┐ ┌──────────────────┐ ┌──────────────────┐     │
│  │ MODULE 1     │ │ MODULE 2     │ │ MODULE 3     │     │
│  │ SECURITY &    │ │ ACADEMIC     │ │ COMMUNICATION │     │
│  │ ANTI-CORRUPTION│ │ MANAGEMENT   │ │ & ENGAGEMENT  │     │
│  │ (10 features) │ │ (8 features) │ │ (5 features) │     │
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│  │ MODULE 4     │ │ MODULE 5     │ │ MODULE 6     │     │
│  │ FINANCE &    │ │ AI &       │ │ INTEGRATION & │     │
│  │ ADMINISTRATION │ │ ANALYTICS    │ │ ACCESSIBILITY │     │
│  │ (6 features) │ │ (4 features) │ │ (5 features) │     │
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│                              │
│  ┌─────────────────────────────────────────────────────────────┐
│  │
│  │ MODULE 7: LIBERIA-SPECIFIC FEATURES (9 features)     │     │
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│
│                              │
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```

MODULE 1: SECURITY & ANTI-CORRUPTION

The foundational layer that makes EK-SMS unique

1.1 Immutable Grade Recording

Description: Once grades are submitted and confirmed, they become permanent records that cannot be altered without creating a visible trail.

Technical Approach:

- Event sourcing architecture (store changes as events, not overwrites)
- Grade states: Draft → Submitted → Locked
- Any modification creates a new event with: who, when, what changed, why

User Stories:

- As a teacher, I can submit grades which become locked after confirmation
- As an admin, I can see the complete history of any grade record
- As a parent, I can trust that my child's grades haven't been tampered with

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1.2 System-Wide Change Alerts

Description: Real-time notifications sent to relevant stakeholders when significant changes occur.

Alert Triggers:

- Grade modification attempt on locked record
- Student enrollment status change
- Fee payment recorded
- Attendance anomaly detected
- User permission changes

Notification Channels:

- In-app notifications (primary)

- Email alerts
- SMS alerts (for critical changes)
- Push notifications (mobile)

1.3 Audit Trail & Forensic Logs

Description: Comprehensive, append-only logging of every action in the system.

What Gets Logged:

- User authentication events (login, logout, failed attempts)
- All CRUD operations on sensitive data
- Permission changes
- Report generation
- Data exports
- System configuration changes

Log Properties:

- Immutable (append-only, no deletions)
- Timestamped with server time
- User identification (who performed action)
- IP address and device info
- Before/after snapshots for data changes

1.4 Blockchain-Backed Record Security (Optional)

Description: Cryptographic verification layer for ultimate tamper-evidence.

Implementation Options:

- Option A: Full blockchain (Hyperledger Fabric for private chain)
- Option B: Merkle tree hashing (simpler, same tamper-evidence)
- Option C: External anchoring (hash records to public blockchain periodically)

Recommendation: Start with Option B (Merkle tree hashing) — provides cryptographic proof without blockchain complexity.

1.5 Two-Factor Authentication (2FA)

Description: Additional authentication layer beyond password.

Methods Supported:

- TOTP (Time-based One-Time Password) via authenticator apps
- SMS OTP (fallback for users without smartphones)
- Email OTP (fallback)

Enforcement:

- Required for: Administrators, Finance staff
- Optional but encouraged for: Teachers, Parents

1.6 Role-Based Access Control (RBAC)

Description: Users can only access features and data permitted by their role.

Core Roles:

```
Super Admin
├── Full system access
├── Can create other admins
└── Audit log access

School Admin
├── School-level configuration
├── User management (within school)
├── Report generation
└── Cannot modify locked grades

Teacher
├── Own classes only
├── Grade entry (own subjects)
├── Attendance marking
└── View own schedule

Student
├── View own grades
├── View own attendance
├── View own fee status
└── Submit feedback

Parent/Guardian
├── View linked student(s) data
├── Receive notifications
├── Pay fees
└── Submit feedback

Finance Officer
├── Fee management
├── Payment processing
├── Financial reports
└── No academic data modification

Examination Officer
├── Exam scheduling
├── Grade compilation
├── Transcript generation
└── Report card generation
```

1.7 Security Audits & Penetration Testing

Description: Regular security assessments to identify vulnerabilities.

Implementation:

- Automated security scanning (OWASP ZAP, Snyk)
- Dependency vulnerability monitoring
- Annual third-party penetration testing
- Security headers and CSP configuration

1.8 Real-Time System Health Monitoring

Description: Continuous monitoring of system security and performance.

Metrics Tracked:

- Failed login attempts (brute force detection)
- Unusual access patterns
- System resource utilization
- Database connection health
- API response times

1.9 End-to-End Encryption

Description: Data protection at rest and in transit.

Implementation:

- TLS 1.3 for all connections
- Database-level encryption (AES-256)
- Encrypted backups
- Secure key management

1.10 User Access Revocation & Lockout

Description: Immediate ability to disable accounts and sessions.

Features:

- Instant account suspension
- Force logout all sessions
- Temporary lockout after failed attempts
- Scheduled access windows (e.g., teacher access only during school hours)

MODULE 2: ACADEMIC MANAGEMENT

2.1 Student Academic Profiles

Description: Centralized, comprehensive student records.

Profile Contents:

- Personal information
- Enrollment history
- Academic records by term/year
- Continuous assessment scores
- Behavioral notes
- Medical/special needs flags
- Guardian information

2.2 Automated Grade Compilation & Calculation

Description: System computes final grades from components.

Features:

- Configurable grading schemes (percentage, letter, GPA)
- Weighted components (exams, coursework, practicals)

- Automatic averaging and ranking
- Grade boundary configuration
- Pass/fail determination

2.3 AI-Powered Timetabling & Scheduling

Description: Intelligent schedule generation considering all constraints.

Constraints Handled:

- Teacher availability
- Room capacity and type requirements
- Subject hour requirements
- Double period requirements
- Break and lunch periods
- Teacher workload balancing

Approach: Agent-based scheduling with validation tools (as discussed previously)

2.4 Report Cards & Transcript Generation

Description: Automated generation of academic documents.

Features:

- Configurable templates
- Digital signatures
- QR code verification
- PDF generation
- Batch generation for entire classes
- Historical transcript compilation

2.5 Attendance Tracking & Analytics

Description: Daily attendance recording with pattern analysis.

Features:

- Multiple attendance states (Present, Absent, Late, Excused)
- Period-by-period or daily tracking
- Absence threshold alerts
- Attendance percentage calculation
- Trend visualization

2.6 Continuous Assessment Tracking

Description: Recording and monitoring of ongoing evaluations.

Features:

- Assignment score recording
- Quiz and test tracking
- Project evaluations
- Practical assessments
- Progress over time visualization
- Early warning for declining performance

2.7 Course & Subject Management

Description: Curriculum structure configuration.

Features:

- Subject creation and categorization
- Credit hour assignment
- Prerequisite mapping
- Teacher-subject assignment

- Class-subject allocation
- Curriculum version control

2.8 Examination Management

Description: End-to-end exam lifecycle management.

Features:

- Exam scheduling
- Room allocation
- Invigilator assignment
- Score entry with verification
- Result compilation
- Grade moderation tools

MODULE 3: COMMUNICATION & ENGAGEMENT

3.1 Parent & Guardian Portal

Description: Dedicated interface for parents to monitor their children.

Access Includes:

- Real-time grade viewing
- Attendance records
- Fee status and payment history
- Upcoming events
- Teacher comments
- Report card downloads

3.2 Secure Internal Messaging

Description: Communication platform within the system.

Features:

- Teacher ↔ Parent messaging
- Admin announcements
- Class-wide messages
- Message read receipts
- Attachment support
- Message archiving

3.3 Push Notifications & SMS/Email Alerts

Description: Multi-channel notification delivery.

Notification Types:

- Grade posted
- Attendance alert
- Fee reminder
- Event announcement
- Emergency broadcast
- System alerts

Delivery Priority:

1. In-app notification
2. Push notification (mobile)
3. Email
4. SMS (critical only, due to cost)

3.4 Anonymous Feedback & Whistleblower Channel

Description: Safe channel for reporting misconduct.

Features:

- Completely anonymous submission
- No tracking of submission origin
- Encrypted content
- Admin-only viewing
- Category-based routing
- Optional follow-up channel (anonymous)

3.5 Event & Announcement Broadcasting

Description: School-wide communication management.

Features:

- Event calendar
- Targeted announcements (by class, grade, role)
- Scheduled publishing
- Attachment support
- Acknowledgment tracking

MODULE 4: FINANCE & ADMINISTRATION

4.1 Cashless Fee Payment System

Description: Digital payment collection and tracking.

Payment Methods:

- Mobile money integration (Orange Money, etc.)
- Bank transfer

- Online card payments
- Local wallet integration

Features:

- Fee structure configuration
- Installment plans
- Late fee calculation
- Scholarship/discount application
- Payment receipts

4.2 Automated Receipt Generation

Description: Digital receipts for all transactions.

Features:

- Unique receipt numbers
- QR code verification
- PDF generation
- Email/SMS delivery
- Receipt lookup portal

4.3 Finance & Budget Dashboards

Description: Financial visibility for administrators.

Metrics:

- Collection rate by term/class
- Outstanding balances
- Revenue vs. budget
- Expense tracking
- Cash flow visualization

4.4 Inventory & Resource Management

Description: Tracking of school assets and supplies.

Items Tracked:

- Textbooks
- Laboratory equipment
- Furniture
- Sports equipment
- Stationery supplies

Features:

- Stock levels
- Assignment to students/rooms
- Condition tracking
- Reorder alerts
- Loss/damage recording

4.5 Staff Payroll & HR Management

Description: Employee management and compensation.

Features:

- Employee profiles
- Contract management
- Salary configuration
- Deduction handling
- Payslip generation
- Leave management
- Performance records

4.6 Document & Certificate Management

Description: Digital archiving of official documents.

Documents Managed:

- Certificates (birth, transfer, etc.)
- Official letters
- Policy documents
- Meeting minutes
- Student submissions

Features:

- Secure storage
- Version control
- Access logging
- Expiry tracking
- Search and retrieval

MODULE 5: AI & ADVANCED ANALYTICS

5.1 Predictive Analytics for Student Performance

Description: AI-powered early warning system.

Predictions:

- At-risk student identification
- Performance trajectory forecasting
- Dropout probability
- Subject-specific struggle detection

Data Inputs:

- Historical grades
- Attendance patterns
- Assessment trends
- Engagement metrics

5.2 Teacher Evaluation Analytics

Description: Data-driven teacher performance insights.

Metrics:

- Class performance averages
- Grade distribution patterns
- Attendance punctuality
- Grading timeliness
- Parent feedback scores
- Peer review (optional)

5.3 Customizable Performance Dashboards

Description: Configurable data visualization for leaders.

Dashboard Types:

- School overview
- Class performance
- Subject analysis
- Teacher comparison
- Trend analysis

5.4 Benchmarking Reports

Description: Comparative analysis across schools.

Comparisons:

- School vs. district average
- Year-over-year trends
- Subject performance ranking
- Attendance benchmarks

MODULE 6: INTEGRATION & ACCESSIBILITY

6.1 Offline Functionality

Description: System works without internet connectivity.

Approach:

- Local database on school premises
- Background sync when connection available
- Conflict resolution for concurrent edits
- Priority sync for critical data (grades)

Technical Options:

- PWA with IndexedDB
- Local server deployment with sync agent
- Hybrid approach

6.2 Multi-Device Access

Description: Accessible from any device type.

Platforms:

- Web application (primary)
- Progressive Web App (installable)
- Mobile responsive design
- Native mobile apps (future phase)
- Desktop application (optional, via Electron)

6.3 Multi-Language Support

Description: Interface in multiple languages.

Languages:

- English (primary)
- Liberian English (localization)
- Local languages (Kpelle, Bassa, etc.)

Implementation:

- i18n framework from day one
- Externalized strings
- RTL support ready

6.4 National Database Integration

Description: Connect with government education systems.

Potential Integrations:

- Ministry of Education student registry
- National examination boards (WAEC)
- National ID systems
- Education statistics reporting

6.5 API Integration Support

Description: Extensibility for third-party connections.

API Design:

- RESTful API architecture
- GraphQL endpoint (optional)
- Webhook support for events
- OAuth2 for third-party authentication
- Rate limiting and API keys

MODULE 7: LIBERIA-SPECIFIC FEATURES

7.1 Anti-Bribery Architecture

Core design principle woven throughout the system

Mechanisms:

- Immutable records eliminate "I can change it for you"
- Multi-party visibility removes secrecy
- Alert system makes tampering obvious
- Audit trails create legal evidence

7.2 Automated Stakeholder Alerting

Description: Proactive notification to all relevant parties.

Alert Recipients:

- Student (owns the record)
- Parent/Guardian (oversight)
- School admin (accountability)
- Ministry (optional, for pattern detection)

7.3 Blockchain-Ready Architecture

Description: Future-proof design for blockchain integration.

Current: Cryptographic hashing of critical records **Future:** Anchor hashes to public/private blockchain

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7.4 Offline-First Design

Description: Primary consideration for low-connectivity areas.

Design Principles:

- Local-first data storage
- Graceful degradation
- Clear online/offline indicators
- Sync status visibility

7.5 National Transparency Focus

Description: System designed for public accountability.

Features:

- Aggregate statistics publishable
- School performance rankings (opt-in)
- Compliance reporting
- Anti-corruption metrics

7.6 Rural Deployment Model

Description: Deployment approach for schools without reliable infrastructure.

Options:

- Local server appliance (Raspberry Pi based)
- Mobile hotspot sync
- Periodic data collection visits
- SMS-based critical operations

7.7 Low-Bandwidth Optimization

Description: Minimal data usage for slow connections.

Optimizations:

- Compressed API responses
- Incremental sync (only changes)
- Image optimization
- Lazy loading
- Offline caching

7.8 Power Outage Resilience

Description: Handling of unreliable electricity.

Features:

- Auto-save on all forms
- Transaction recovery
- UPS recommendations for servers
- Battery backup for local installations

7.9 Local Currency & Context

Description: Liberian-specific configurations.

Includes:

- Liberian Dollar (LRD) currency
- Local date formats
- Local phone number formats
- Liberian school calendar integration
- Public holiday awareness

PART 2: PHASED IMPLEMENTATION PLAN

Phase Overview

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┌──────────────────────────────────────────────────────────────────────
│                IMPLEMENTATION PHASES                           │
├──────────────────────────────────────────────────────────────────────
│                                              │
│ PHASE 1: MVP (Core Anti-Corruption)        Timeline: 3-4 months      │
│ ─────────────────────────────────────────────                  │
│ • User authentication & RBAC                         │
│ • Student management                              │
│ • Immutable grade recording                         │
│ • Audit trail                                   │
│ • Basic alerts                                  │
│ • Parent portal (read-only)                       │
│ • Report card generation                          │
│                                              │
│ PHASE 2: Academic Expansion             Timeline: 2-3 months      │
│ ─────────────────────────────────────────                  │
│ • Attendance tracking                              │
│ • Continuous assessment                             │
│ • Course/subject management                          │
│ • Full messaging system                            │
│ • Enhanced dashboards                              │
│ • Whistleblower channel                            │
│                                              │
│ PHASE 3: Finance & Administration        Timeline: 2-3 months      │
│ ─────────────────────────────────────────                  │
│ • Fee tracking & payment integration                   │
│ • Receipt generation                             │
│ • Finance dashboards                              │
```

```
│ • Inventory management                    │
│ • Document management                     │
│                                           │
│ PHASE 4: Intelligence Layer       Timeline: 2-3 months   │
│ ─────────────────────────────                            │
│ • AI timetabling agent                    │
│ • Predictive analytics                    │
│ • Teacher evaluation                      │
│ • Advanced dashboards                     │
│ • Benchmarking reports                    │
│                                           │
│ PHASE 5: Offline & Scale          Timeline: 3-4 months   │
│ ──────────────────────────                               │
│ • Offline-first architecture              │
│ • Local server deployment                 │
│ • Sync mechanism                          │
│ • National database integration           │
│ • Multi-language support                  │
│ • Blockchain anchoring                    │
│                                           │
│ PHASE 6: HR & Payroll             Timeline: 2 months     │
│ ───────────────────────                                  │
│ • Staff management                        │
│ • Payroll processing                      │
│ • Leave management                        │
│ • Performance tracking                    │
│                                           │
└──────────────────────────────────────────────────────────

Total Estimated Timeline: 14-19 months for complete system
```

PART 3: MVP SPECIFICATION

MVP Philosophy

The MVP focuses on the **core value proposition**: **A transparent, tamper-proof grade management system that restores trust in academic records.**

Everything else can be layered on once this foundation proves itself.

MVP Feature Set

Authentication & Access Control ✓

Feature	Included	Notes
User registration	✓	Admin-controlled
Login/logout	✓	
Password reset	✓	Email-based
Two-factor authentication	✓	TOTP + SMS fallback
Role-based access	✓	5 core roles
Session management	✓	

User Roles in MVP

```
┌─────────────────────────────────────────────────────────────┐
│                    MVP ROLES                                │
├─────────────────────────────────────────────────────────────┤
│                                                             │
│  ADMIN                                                      │
│  ├── User management                                        │
│  ├── School configuration                                   │
│  ├── View all data                                          │
│  ├── Audit log access                                       │
│  └── Report generation                                      │
│                                                             │
│  TEACHER                                                    │
│  ├── View assigned classes                                  │
│  ├── Enter grades (own subjects)                            │
│  ├── Submit grades for locking                              │
│  └── View student profiles (own classes)                    │
│                                                             │
│  STUDENT                                                    │
│  ├── View own grades                                        │
│  ├── View own profile                                       │
│  └── Download report cards                                  │
│                                                             │
```

```
│  PARENT                                         │
│  ├── View linked student(s) grades              │
│  ├── Receive notifications                      │
│  └── Download report cards                      │
│                                                 │
│  EXAMINATION OFFICER                            │
│  ├── Compile grades                             │
│  ├── Generate report cards                      │
│  └── Generate transcripts                       │
│                                                 │
└─────────────────────────────────────────────────┘
```

Academic Core ✓

Feature	Included	Notes
Student profiles	✓	Basic info + academic history
Class management	✓	Class creation, student assignment
Subject management	✓	Subject creation, teacher assignment
Term/semester configuration	✓	Academic calendar
Grade entry	✓	By teacher, per subject
Grade locking	✓	Immutable after confirmation
Grade calculation	✓	Weighted averages
Class rankings	✓	
Report card generation	✓	PDF with QR verification

Anti-Corruption Core ✓

Feature	Included	Notes
Immutable grade recording	✓	Event sourcing

Feature	Included	Notes
Change attempt alerts	✓	Email + in-app
Complete audit trail	✓	All actions logged
Cryptographic hashing	✓	Merkle tree for records
Grade verification	✓	QR code on documents

Parent Portal ✓

Feature	Included	Notes
View child's grades	✓	Real-time
View academic history	✓	All terms
Download report cards	✓	PDF
Notifications	✓	Grade posted, alerts

Notifications ✓

Feature	Included	Notes
In-app notifications	✓	Primary channel
Email notifications	✓	All users
SMS notifications	Partial	Critical alerts only

NOT in MVP (Deferred)

Feature	Phase	Reason
Attendance tracking	2	Not core to anti-corruption mission

Feature	Phase	Reason
Continuous assessment	2	Enhancement, not essential
Messaging system	2	Adds complexity
Fee management	3	Requires payment integration
Inventory	3	Administrative, not core
AI timetabling	4	Complex, needs stable base
Predictive analytics	4	Needs historical data
Offline functionality	5	Major architectural effort
Payroll	6	Separate domain
Blockchain	5	Merkle trees sufficient initially

MVP User Flows

Flow 1: Admin Sets Up School

```
1. Admin logs in with 2FA
2. Admin creates academic year and terms
3. Admin creates classes (Grade 10A, Grade 10B, etc.)
4. Admin creates subjects
5. Admin creates teacher accounts
6. Admin assigns teachers to subjects and classes
7. Admin imports or creates student records
8. Admin creates parent accounts and links to students
```

Flow 2: Teacher Enters Grades

```
1. Teacher logs in with 2FA
2. Teacher sees assigned classes
3. Teacher selects class and subject
4. Teacher sees list of students
5. Teacher enters grades for each student
6. System auto-saves as draft
7. Teacher reviews and clicks "Submit for Locking"
```

```
8. System confirms: "These grades will be permanently recorded"
9. Teacher confirms
10. Grades become immutable
11. Notifications sent to students and parents
12. Audit log records the submission
```

Flow 3: Parent Checks Grades

```
1. Parent logs in
2. Parent sees linked student(s)
3. Parent views current term grades
4. Parent sees subject-by-subject breakdown
5. Parent downloads report card PDF
6. Parent scans QR code to verify authenticity
```

Flow 4: Grade Modification Attempt

```
1. Teacher or admin attempts to modify locked grade
2. System blocks direct modification
3. System requires: reason, supporting evidence
4. Modification request created (not applied)
5. Alert sent to: student, parent, admin, audit log
6. Admin reviews modification request
7. If approved: new grade event added (old grade preserved)
8. If rejected: request denied, logged
9. Full history visible in audit trail
```

Flow 5: Report Card Generation

```
1. Exam officer logs in
2. Selects term and class
3. System compiles all grades
4. System calculates averages, rankings
5. Officer reviews compilation
6. Officer generates report cards (batch)
7. Each report card includes:
   - Cryptographic hash
   - QR code linking to verification
   - Digital signature
```

```
8. Report cards available for download
9. Notification sent to parents
```

MVP Data Model

```
┌──────────────────────────────────────────────────────────────────────────────
│                    MVP DATA MODEL                          │
├──────────────────────────────────────────────────────────────────────────────
│                                                            │
│ CORE ENTITIES                                              │
│ ─────────────                                              │
│                                                            │
│ School                                                     │
│ ├── id, name, code                                         │
│ ├── address, contact                                       │
│ └── settings (grading scheme, term structure)              │
│                                                            │
│ User                                                       │
│ ├── id, email, phone                                       │
│ ├── password_hash                                          │
│ ├── role (admin, teacher, student, parent, exam_officer)   │
│ ├── two_factor_secret                                      │
│ └── school_id                                              │
│                                                            │
│ Student                                                    │
│ ├── id, user_id                                            │
│ ├── student_number                                         │
│ ├── first_name, last_name, date_of_birth                   │
│ ├── gender                                                 │
│ ├── enrollment_date                                        │
│ ├── current_class_id                                       │
│ └── status (active, graduated, transferred)                │
│                                                            │
│ Parent                                                     │
│ ├── id, user_id                                            │
│ ├── first_name, last_name                                  │
│ └── relationship (father, mother, guardian)                │
│                                                            │
│ StudentParent (junction)                                   │
│ ├── student_id                                             │
│ └── parent_id                                              │
```

```
│                                              │
│ Teacher                                      │
│ ├── id, user_id                              │
│ ├── employee_number                          │
│ ├── first_name, last_name                    │
│ └── specializations                          │
│                                              │
│ AcademicYear                                 │
│ ├── id, school_id                            │
│ ├── name (e.g., "2024-2025")                 │
│ ├── start_date, end_date                     │
│ └── is_current                               │
│                                              │
│ Term                                         │
│ ├── id, academic_year_id                     │
│ ├── name (e.g., "First Term")                │
│ ├── start_date, end_date                     │
│ └── is_current                               │
│                                              │
│ Class                                        │
│ ├── id, school_id                            │
│ ├── name (e.g., "Grade 10A")                 │
│ ├── grade_level                              │
│ └── academic_year_id                         │
│                                              │
│ Subject                                      │
│ ├── id, school_id                            │
│ ├── name, code                               │
│ ├── category (core, elective)                │
│ └── credit_hours                             │
│                                              │
│ ClassSubject (what subjects a class takes)   │
│ ├── id, class_id, subject_id                 │
│ ├── teacher_id                               │
│ └── term_id                                  │
│                                              │
│ StudentClass (enrollment)                    │
│ ├── student_id, class_id                     │
│ └── enrolled_at                              │
│                                              │
│                                              │
│ GRADE MANAGEMENT (Event-Sourced)             │
│ ──────────────────────────────────────       │
│                                              │
```

```
│ GradeEvent (immutable, append-only)                                    │
│ ├── id (UUID)                                                          │
│ ├── student_id                                                         │
│ ├── class_subject_id                                                   │
│ ├── term_id                                                            │
│ ├── event_type (DRAFT, SUBMIT, LOCK, MODIFY_REQUEST, MODIFY_APPROVED)  │
│ ├── score                                                              │
│ ├── grade_letter                                                       │
│ ├── remarks                                                            │
│ ├── recorded_by (user_id)                                              │
│ ├── recorded_at (timestamp)                                            │
│ ├── modification_reason (nullable)                                     │
│ ├── previous_event_id (for chain)                                      │
│ └── hash (cryptographic, includes previous_hash)                       │
│                                                                        │
│ CurrentGrade (materialized view for easy querying)                     │
│ ├── student_id                                                         │
│ ├── class_subject_id                                                   │
│ ├── term_id                                                            │
│ ├── current_score                                                      │
│ ├── current_grade_letter                                               │
│ ├── status (draft, locked)                                             │
│ └── last_event_id                                                      │
│                                                                        │
│                                                                        │
│ AUDIT & SECURITY                                                       │
│ ────────────────────────                                               │
│                                                                        │
│ AuditLog (append-only)                                                 │
│ ├── id                                                                 │
│ ├── user_id                                                            │
│ ├── action (LOGIN, LOGOUT, VIEW, CREATE, UPDATE, DELETE, SUBMIT, LOCK) │
│ ├── entity_type (Student, Grade, Class, etc.)                          │
│ ├── entity_id                                                          │
│ ├── before_state (JSON)                                                │
│ ├── after_state (JSON)                                                 │
│ ├── ip_address                                                         │
│ ├── user_agent                                                         │
│ ├── timestamp                                                          │
│ └── hash (includes previous log hash)                                  │
│                                                                        │
│ Notification                                                           │
│ ├── id                                                                 │
│ ├── user_id                                                            │
```

```
│  ├── type (GRADE_POSTED, GRADE_LOCKED, MODIFICATION_ATTEMPT, etc.)  │
│  ├── title                                   │
│  ├── message                                 │
│  ├── related_entity_type                     │
│  ├── related_entity_id                       │
│  ├── is_read                                 │
│  ├── sent_email                              │
│  ├── sent_sms                                │
│  └── created_at                              │
│                                              │
│ Session                                      │
│  ├── id                                      │
│  ├── user_id                                 │
│  ├── token_hash                              │
│  ├── ip_address                              │
│  ├── user_agent                              │
│  ├── created_at                              │
│  ├── expires_at                              │
│  └── is_revoked                              │
│                                              │
│                                              │
│ DOCUMENTS                                    │
│ ─────────                                    │
│                                              │
│ ReportCard                                   │
│  ├── id                                      │
│  ├── student_id                              │
│  ├── term_id                                 │
│  ├── generated_at                            │
│  ├── generated_by (user_id)                  │
│  ├── file_path                               │
│  ├── verification_hash                       │
│  └── qr_code                                 │
│                                              │
└──────────────────────────────────────────────
```

MVP Technology Stack (Recommendation)

```
┌──────────────────────────────────────────────
│                MVP TECH STACK                │
```

FRONTEND

- Next.js 14+ (App Router)
- TypeScript
- Tailwind CSS
- shadcn/ui components
- React Query for data fetching
- Zustand for client state

BACKEND

- Python / FastAPI
- SQLAlchemy 2.0 (async)
- Pydantic for validation
- Alembic for migrations

DATABASE

- PostgreSQL (primary)
- Redis (sessions, caching)

INFRASTRUCTURE

- Docker / Docker Compose
- Nginx (reverse proxy)
- Let's Encrypt (SSL)

SERVICES

- Email: Resend or SendGrid
- SMS: Twilio or local provider
- PDF Generation: WeasyPrint or ReportLab
- QR Codes: python-qrcode

SECURITY

- Argon2 for password hashing
- PyOTP for TOTP
- PyJWT for tokens
- Cryptography library for hashing

```
│                                            │
└──────────────────────────────────────────────────────────────────────────────────────────
```

MVP API Structure

```
/api/v1
├── /auth
│   ├── POST   /register        (admin creates users)
│   ├── POST   /login
│   ├── POST   /logout
│   ├── POST   /refresh
│   ├── POST   /2fa/setup
│   ├── POST   /2fa/verify
│   └── POST   /password/reset
│
├── /users
│   ├── GET    /me
│   ├── PATCH  /me
│   └── GET    /:id             (admin only)
│
├── /schools
│   ├── GET    /current
│   └── PATCH  /current         (admin only)
│
├── /academic-years
│   ├── GET    /
│   ├── POST   /                (admin only)
│   └── GET    /:id
│
├── /terms
│   ├── GET    /
│   ├── POST   /                (admin only)
│   └── GET    /:id
│
├── /classes
│   ├── GET    /
│   ├── POST   /                (admin only)
│   ├── GET    /:id
│   ├── GET    /:id/students
│   └── GET    /:id/subjects
```

```
│
├── /subjects
│   ├── GET    /
│   ├── POST   /                (admin only)
│   └── GET    /:id
│
├── /students
│   ├── GET    /
│   ├── POST   /                (admin only)
│   ├── GET    /:id
│   ├── GET    /:id/grades
│   └── GET    /:id/report-cards
│
├── /teachers
│   ├── GET    /
│   ├── GET    /:id
│   └── GET    /:id/assignments
│
├── /grades
│   ├── GET    /class/:classId/subject/:subjectId/term/:termId
│   ├── POST   /                (teacher: save draft)
│   ├── POST   /submit          (teacher: submit for locking)
│   ├── POST   /lock            (confirm lock)
│   ├── GET    /:id/history     (audit trail)
│   └── POST   /:id/modify-request (request modification)
│
├── /report-cards
│   ├── POST   /generate        (exam officer)
│   ├── GET    /:id
│   ├── GET    /:id/download
│   └── GET    /verify/:hash    (public verification)
│
├── /notifications
│   ├── GET    /
│   ├── PATCH  /:id/read
│   └── POST   /read-all
│
└── /audit-logs
    ├── GET    /                (admin only)
    └── GET    /entity/:type/:id  (admin only)
```

MVP Screens

Admin Screens

1. Dashboard (overview stats)
2. User Management (CRUD for all user types)
3. Academic Year & Term Management
4. Class Management
5. Subject Management
6. Teacher Assignments
7. Student Enrollment
8. Audit Log Viewer
9. School Settings

Teacher Screens

1. Dashboard (assigned classes)
2. Class View (students in class)
3. Grade Entry (per subject, per term)
4. Grade Submission Confirmation
5. Grade History View (own submissions)

Student Screens

1. Dashboard (current grades summary)
2. Grades by Term
3. Report Card Download
4. Notification Center

Parent Screens

1. Dashboard (children overview)
2. Child Grades View
3. Report Card Download

4. Notification Center

Exam Officer Screens

1. Dashboard
2. Grade Compilation View
3. Report Card Generation
4. Verification History

MVP Timeline Estimate

```
Week 1-2:   Project setup, database schema, auth system
Week 3-4:   Core models (School, User, Student, Teacher, Class, Subject)
Week 5-6:   Grade event system with immutability
Week 7-8:   Teacher grade entry flow + locking mechanism
Week 9-10:  Audit logging + notification system
Week 11-12: Parent portal + report card generation
Week 13-14: Testing, security hardening, documentation
Week 15-16: Deployment, pilot testing, bug fixes

Total: ~4 months for solo developer working full-time
```

MVP Success Criteria

Technical

- [ ] 2FA enforced for all admin/teacher accounts
- [ ] No grade modification possible without audit trail
- [ ] All actions logged immutably
- [ ] Report cards include verifiable QR codes
- [ ] Parent portal shows real-time grades

Functional

- [ ] Admin can set up complete school structure
- [ ] Teachers can enter and lock grades

- [ ] Students and parents can view grades
- [ ] Report cards can be generated and verified
- [ ] Modification attempts trigger alerts

Security

- [ ] Penetration test passed
- [ ] OWASP Top 10 vulnerabilities addressed
- [ ] Data encrypted at rest and in transit
- [ ] Session management secure

Performance

- [ ] Page load < 3 seconds on 3G
- [ ] API response < 500ms average
- [ ] Supports 100 concurrent users

Post-MVP Roadmap Summary

Phase	Focus	Key Features	Timeline
2	Academic Depth	Attendance, continuous assessment, messaging	2-3 months
3	Finance	Fee management, payments, receipts	2-3 months
4	Intelligence	AI timetabling, analytics, predictions	2-3 months
5	Scale & Offline	Offline-first, sync, multi-language	3-4 months
6	HR	Payroll, staff management	2 months

Appendix: Feature-to-Phase Mapping

Feature	Module	Phase
Immutable grade recording	Security	MVP
System-wide change alerts	Security	MVP

Feature	**Module**	**Phase**
Audit trail & forensic logs	Security	MVP
Blockchain-backed security	Security	5
Two-factor authentication	Security	MVP
Role-based access control	Security	MVP
Penetration testing	Security	MVP+
System health monitoring	Security	2
End-to-end encryption	Security	MVP
Access revocation & lockout	Security	MVP
Student academic profiles	Academic	MVP
Grade compilation & calculation	Academic	MVP
AI-powered timetabling	Academic	4
Report cards & transcripts	Academic	MVP
Attendance tracking	Academic	2
Continuous assessment	Academic	2
Course & subject management	Academic	MVP
Examination management	Academic	2
Parent & guardian portal	Communication	MVP
Secure internal messaging	Communication	2
Push/SMS/Email alerts	Communication	MVP
Anonymous whistleblower	Communication	2
Event broadcasting	Communication	2
Cashless fee payment	Finance	3
Receipt generation	Finance	3

Feature	Module	Phase
Finance dashboards	Finance	3
Inventory management	Finance	3
Staff payroll & HR	Finance	6
Document management	Finance	3
Predictive analytics	AI	4
Teacher evaluation analytics	AI	4
Performance dashboards	AI	4
Benchmarking reports	AI	4
Offline functionality	Integration	5
Multi-device access	Integration	MVP
Multi-language support	Integration	5
National database integration	Integration	5
API integration support	Integration	MVP

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